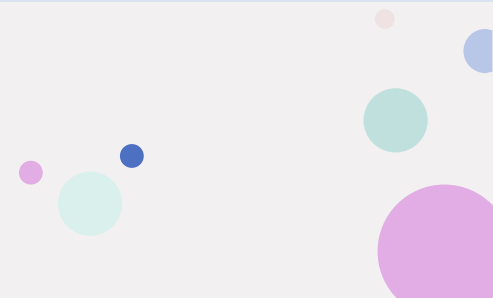




Lecture 2: Perfect Secure Encryption

CS409: Introduction to Cryptography
Instructor: Sruthi Sekar



Today's Class



Perfect Secure
Encryption: how do
we define this
formally?



Build perfect secure
encryption – Vernam
Cipher



Rigorous security
proof

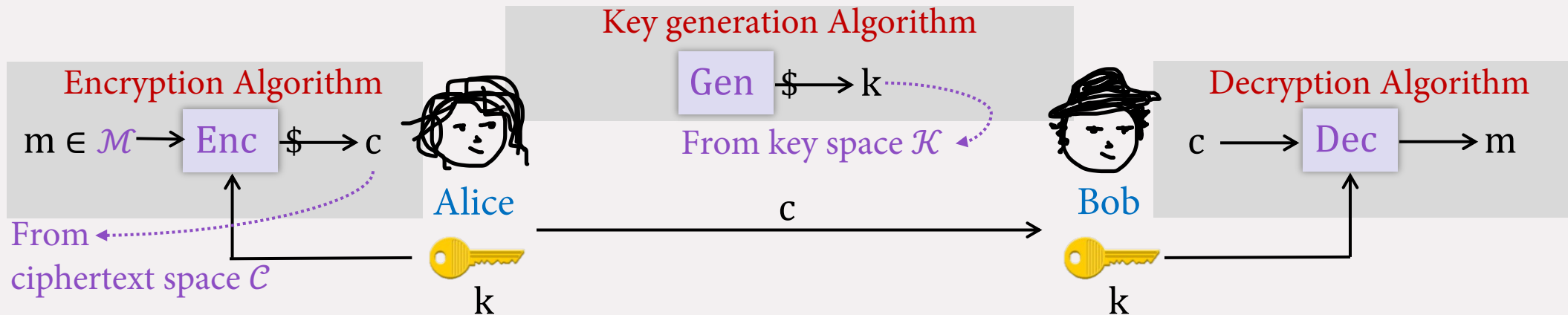


Drawbacks, Attacks

Symmetric Key Encryption (SKE): Syntax

Definition 1 (Symmetric Key Encryption (SKE)):

An SKE for message space $\mathcal{M}$ is a triple of efficient algorithms $(\text{Gen}, \text{Enc}, \text{Dec})$ with syntax:



- Correctness of decryption:** For all messages $m \in \mathcal{M}$,
$$\Pr_{k \leftarrow \text{Gen}, c \leftarrow \text{Enc}(k, m)} [\text{Dec}(k, c) = m] = 1$$



While Gen and Enc are randomized algorithms, Dec is a deterministic algorithm. Why?

**** $y \leftarrow A(x)$:** denotes running randomized A on input x and getting (random) y

Formally Defining Security

THREAT MODEL

How powerful is Eve (computationally)?

Unbounded computation power

How can Eve attack the communication?

Passive or Eavesdropper
Ciphertext Only Attack (COA) 

Eve deterministic or randomized?

Randomized
(more powerful than deterministic)



Eve

BREAK MODEL

Attempt I: Eve can find the key?

Then $\text{Enc}(m) = m$ is secure. 

Attempt II: Eve can recover entire m

$\text{Enc}(m)$ leaking most significant 10 bits
(secure, but what if your password?) 

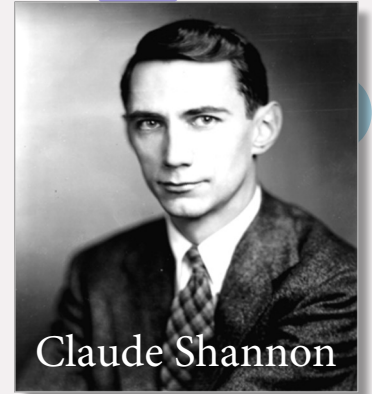
Shannon's Idea

C should reveal no information about m 

Let's formalize this!

Perfect Security: Formal Definition

[Shannon 1949]



Definition 2 (Perfect Security):

An encryption scheme (Gen, Enc, Dec) with message space $\mathcal{M}$ is **perfectly secure** if for every probability distribution M over $\mathcal{M}$, every message $m \in \mathcal{M}$, and every ciphertext $c \in \mathcal{C}$ for which $\Pr[C = c] > 0$:

$$\Pr[M = m | C = c] = \Pr[M = m]$$

Posterior probability
that m is encrypted in c

A priori probability that m
was the message picked.



Eve

Any information that Eve had about the message, before and after seeing the ciphertext, should be the same.



** C : denotes the ciphertext distribution induced by M , Gen and Enc.

Perfect Security: Alternate Definition

Definition 3

An encryption scheme (Gen, Enc, Dec) with message space $\mathcal{M}$ is **perfectly secure** if for every probability distribution over $\mathcal{M}$, every $m_0, m_1 \in \mathcal{M}$, and every ciphertext $c \in \mathcal{C}$:

$$\Pr[C = c \mid M = m_0] = \Pr[C = c \mid M = m_1]$$
$$\Pr_{k \leftarrow \text{Gen}} [\text{Enc}_k(m_0) = c] = \Pr_{k' \leftarrow \text{Gen}} [\text{Enc}_{k'}(m_1) = c]$$



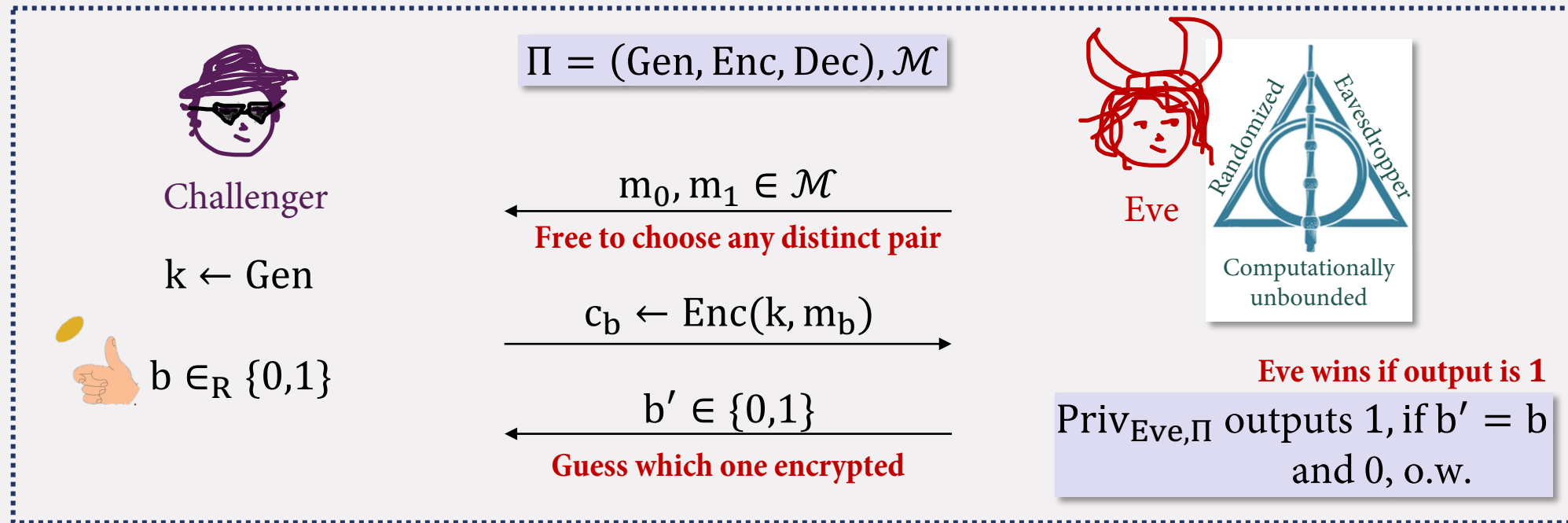
Eve

Distribution of ciphertext is independent of plaintext

For two messages m_0 and m_1 , the distribution of ciphertext when m_0 was encrypted should be identical to its distribution when m_1 was encrypted!



Perfect Security: Indistinguishability-based



Definition 4 (Perfect Indistinguishability):

An encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ is **perfectly indistinguishable** if for every adversary **Eve**, it holds that:

$$\Pr[\text{Priv}_{\text{Eve}, \Pi} = 1] = 1/2$$

Best Eve can do is randomly guess



** $x \in_R \mathcal{S}$: denotes picking b uniformly at random from the set $\mathcal{S}$

Questions

Shift Cipher:

- **Gen**: Pick $k \in_R \{0, 1, \dots, 25\}$
- **Enc**($k, m = m_1 \dots m_\ell$) $:= c_1 \dots c_\ell$, where $c_i = m_i + k \bmod 26$
- **Dec**($k, c = c_1 \dots c_\ell$) $:= m_1 \dots m_\ell$, where $m_i = c_i - k \bmod 26$

HW: Is the shift cipher perfectly secure?

Substitution Cipher:

- **Gen**: Pick $k \in_R \{f: \{0, 1, \dots, 25\} \rightarrow \{0, 1, \dots, 25\}: f \text{ is a permutation}\}$
- **Enc**(k, m) $:= k(m)$
- **Dec**(k, c) $:= k^{-1}(c)$

HW: Is the substitution cipher perfectly secure/indistinguishable?

Today's Class



Perfect Secure
Encryption: how do
we define this
formally?



Build perfect
secure encryption
(Vernam Cipher)



Rigorous proof
of security



Drawbacks, Attacks

Vernam Cipher (One-time pad) [1917]



Mid-20th century

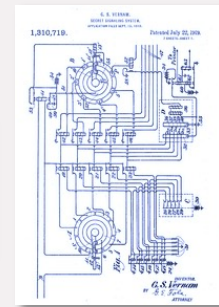
Red phone: linked White House and Kremlin during Cold War

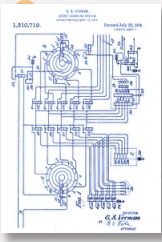


Long keys were shared using trusted couriers carrying briefcases of paper with random characters

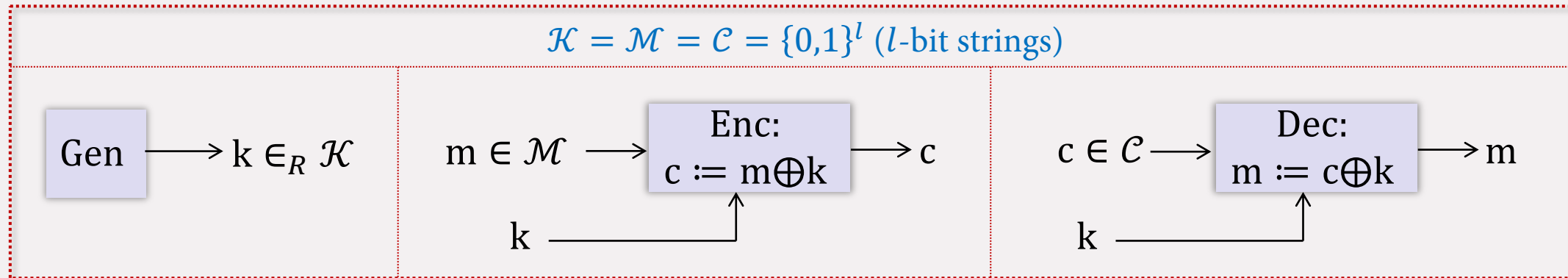


About 25 years later, Shannon showed that the Vernam cipher is actually perfectly secure!





Vernam Cipher (One-time pad) [1917]



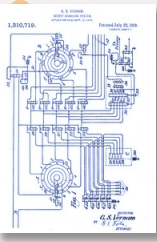
Correctness

$$\text{Dec}(k, \text{Enc}(k, m)) = k \oplus (k \oplus m) = m$$

Theorem 1 (Security of Vernam Cipher):

One-time pad (OTP) is a perfect secure encryption.

** $\oplus$: denotes the XOR operation, which is addition modulo 2



Vernam Cipher (One-time pad) [1917]



Theorem 1 (Security of Vernam Cipher):

One-time pad (OTP) is a perfect secure encryption.

Proof:

To prove: For every distribution M on $\mathcal{M}$, $\forall m \in \mathcal{M}$, $\forall c \in \mathcal{C}$ ($\Pr[C = c] > 0$),
 $\Pr[M = m | C = c] = \Pr[M = m]$

$$\Pr[M = m | C = c] = \frac{\Pr[C = c | M = m] \cdot \Pr[M = m]}{\Pr[C = c]} \quad (\text{Bayes' Theorem})$$

(For any $m \in \mathcal{M}$)

$$\begin{aligned} \Pr[C = c | M = m] &= \Pr[\text{Enc}_k(m) = c] \\ &= \Pr[m \oplus k = c] \\ &= \Pr[k = m \oplus c] \\ &= \frac{1}{2^\ell} \end{aligned}$$

$$\begin{aligned} \Pr[C = c] &= \sum_{m' \in \mathcal{M}} \Pr[C = c | M = m'] \cdot \Pr[M = m'] \\ &= \frac{1}{2^\ell} \sum_{m' \in \mathcal{M}} \Pr[M = m'] = \frac{1}{2^\ell} \end{aligned}$$

$$\therefore \Pr[M = m | C = c] = \Pr[M = m]$$

Today's Class



Perfect Secure
Encryption: how do
we define this
formally?



Build perfect
secure encryption
(Vernam Cipher)

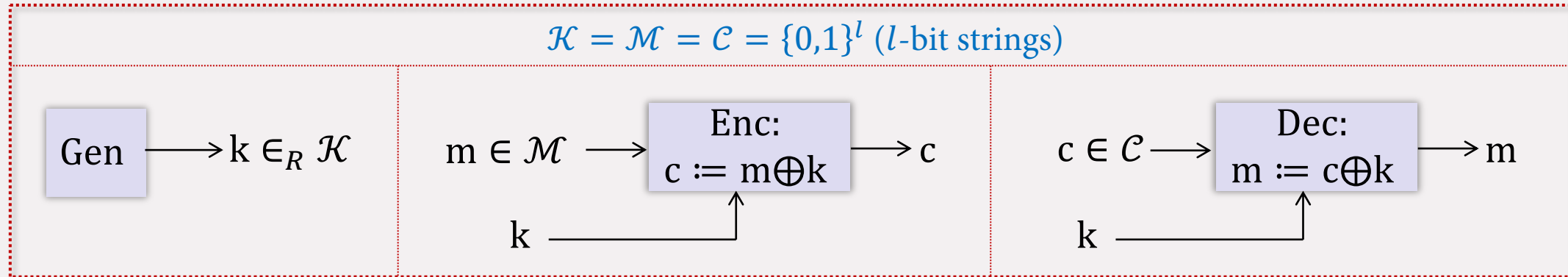


Rigorous proof
of security



Drawbacks, Attacks

Vernam Cipher: Drawbacks



1. How long is the key?

As long as the message

2. Can you at least reuse the key?

No, two-time pad insecure!

Some other scheme maybe?



$$c = m \oplus k$$

 $\oplus$

$$c' = m' \oplus k$$

 $=$

$$m' \oplus m$$

VENONA Project: US and UK decrypted ciphertexts by Soviet Union that were encrypted with same key!

***Wait for the Lab exercise 1!!**

Drawback of Perfect Secure Encryption

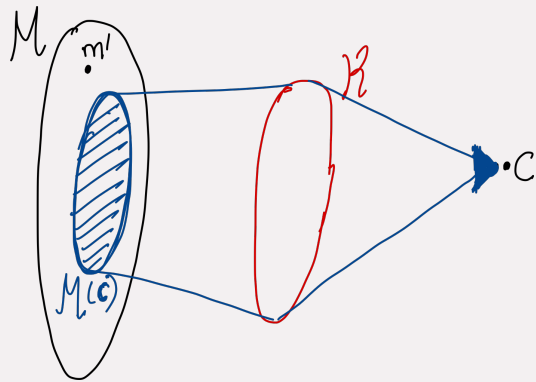
Theorem 2

If $(\text{Gen}, \text{Enc}, \text{Dec})$ is a perfect secure encryption with message space $\mathcal{M}$ and key space $\mathcal{K}$, then $|\mathcal{K}| \geq |\mathcal{M}|$.

Proof:

Assume to the contrary that $|\mathcal{K}| < |\mathcal{M}|$. We will contradict the perfect security.

Let $c \in \mathcal{C}$ be such that $\Pr[C = c] > 0$. Consider $\mathcal{M}(c)$ to be the messages that are decryptions of c .



$$\mathcal{M}(c) := \{m \mid m = \text{Dec}(k, c) \text{ for some } k \in \mathcal{K}\}$$

$|\mathcal{M}(c)| \leq |\mathcal{K}|$ (Dec is deterministic). Now, since we assumed $|\mathcal{K}| < |\mathcal{M}|$, there must be some $m' \in \mathcal{M}$ such that $m' \notin \mathcal{M}(c)$.

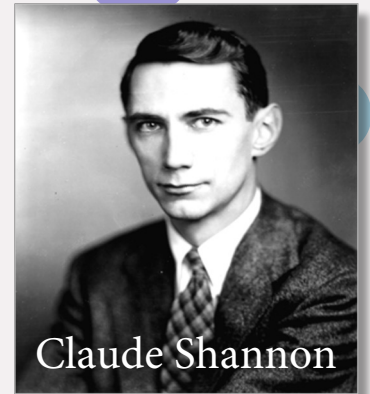
$$\Rightarrow \Pr[M = m' | C = c] = 0 \neq \underbrace{\Pr[M = m']}_{\text{Take } M \text{ to be such a distribution}}$$

Take M to be such a distribution



First drawback of OTP inherent to perfect security!

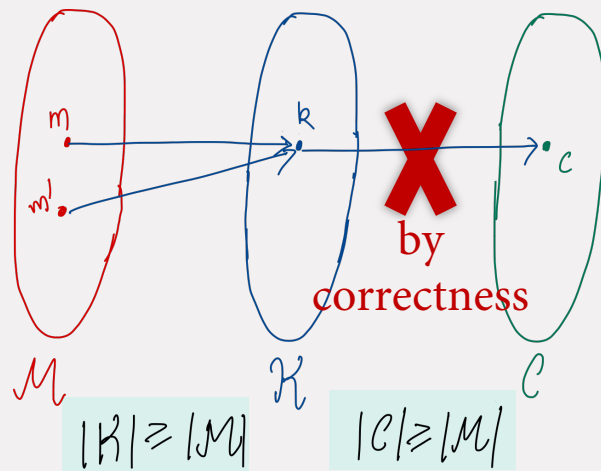
Perfect Security: Shannon's Theorem



Definition 5: Shannon's Theorem

An encryption scheme (Gen, Enc, Dec) with message space $\mathcal{M}$ such that $|\mathcal{M}| = |\mathcal{K}| = |\mathcal{C}|$ is **perfectly secure** if and only if:

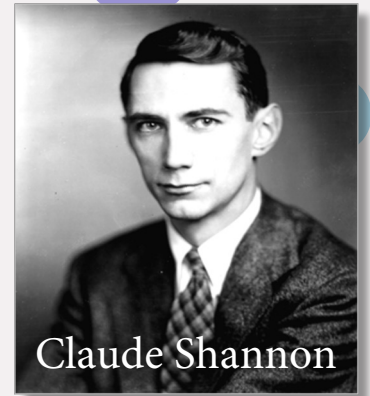
1. Every $k \in \mathcal{K}$ is chosen with (equal) probability $1/|\mathcal{K}|$ by Gen.
2. For every $m \in \mathcal{M}$ and every $c \in \mathcal{C}$, there exists a **unique key** $k \in \mathcal{K}$ such that $\text{Enc}(k, m)$ outputs c .



Correctness $\Rightarrow |\mathcal{C}| \geq |\mathcal{M}|$
 $|\mathcal{M}| = |\mathcal{K}| = |\mathcal{C}|$ is optimal choice!

Second drawback of OTP also inherent to perfect security!

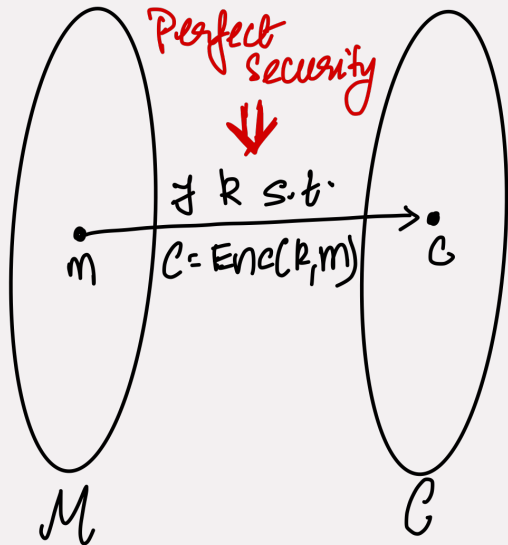
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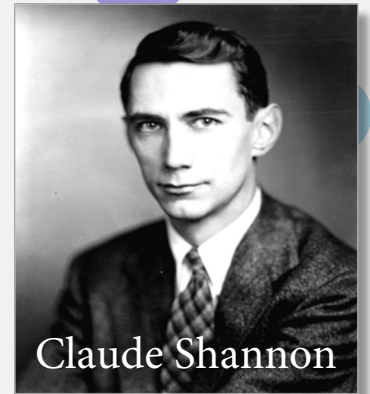
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$|\mathcal{M}| = |\mathcal{K}| = |\mathcal{C}| \Rightarrow$ this k is unique!

Second drawback of OTP also inherent to perfect security!

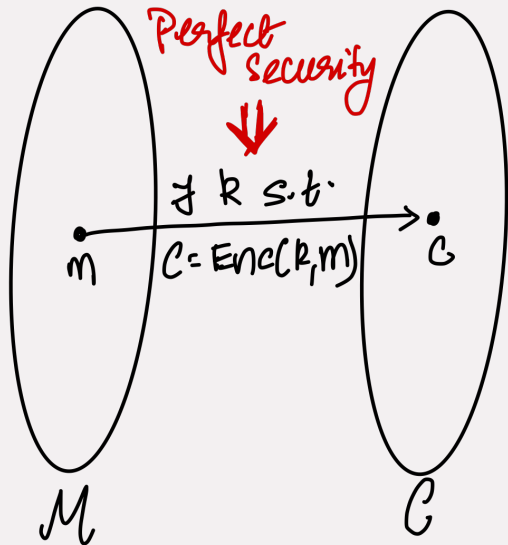
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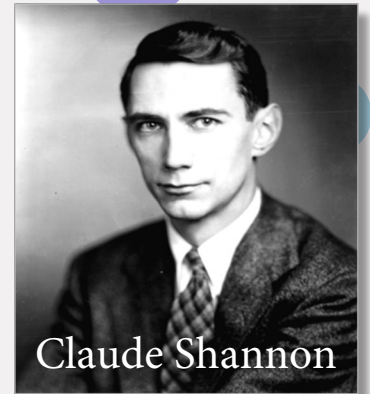


$|\mathcal{M}| = |\mathcal{K}| = |\mathcal{C}| \Rightarrow$ this k is unique!

Perfect security $\Rightarrow$ Each key chosen with equal probability!

Second drawback of OTP also inherent to perfect security!

Perfect Security: Shannon's Theorem



Definition 5: Shannon's Theorem

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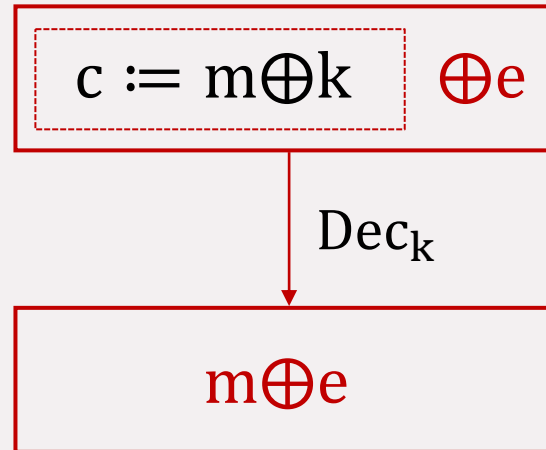
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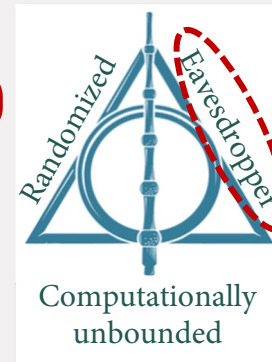
HW: Show the equivalence of all four definitions: $\text{Def 2} \Leftrightarrow \text{Def 3} \Leftrightarrow \text{Def 4} \Leftrightarrow \text{Def 5}$

Vernam Cipher: Malleability Attacks

What if Eve could modify messages being sent?



Eve



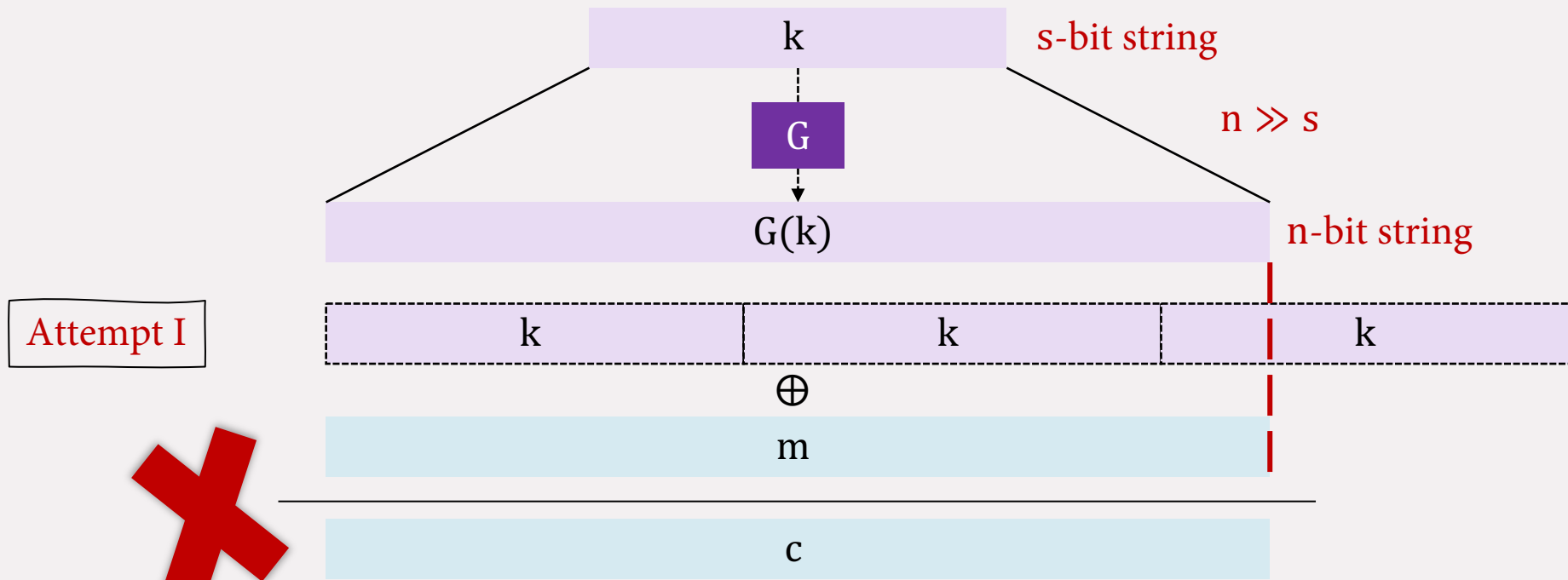
We will later see how to strengthen the threat model and build encryption schemes with integrity!

NO INTEGRITY

No way to detect you received the actual message or not!

How to overcome these limitations?

First limitation: Key as long as message! Let's try and fix using some process that expands the key..

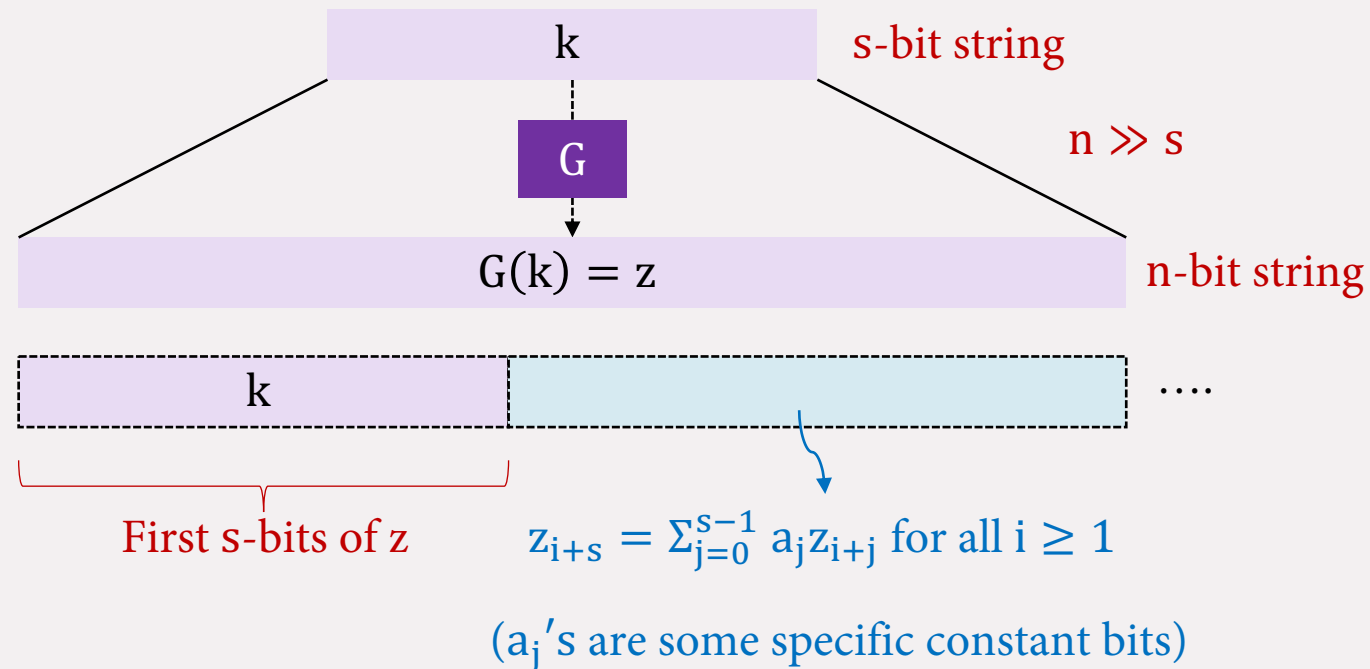


WHERE DID YOU SEE THIS? Vignère Cipher!

How to overcome these limitations?

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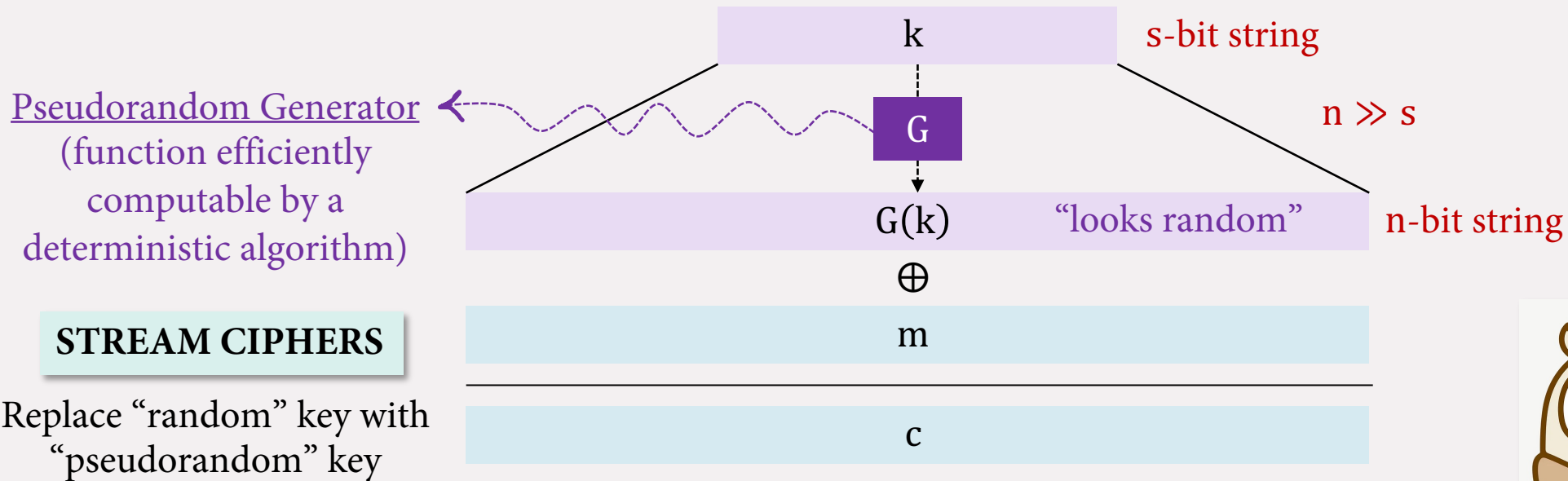
Attempt II



IS THIS GOOD ENOUGH?

How to overcome these limitations?

First limitation: Key as long as message! Let's try and fix using some process that expands the key..



NEXT
CLASS

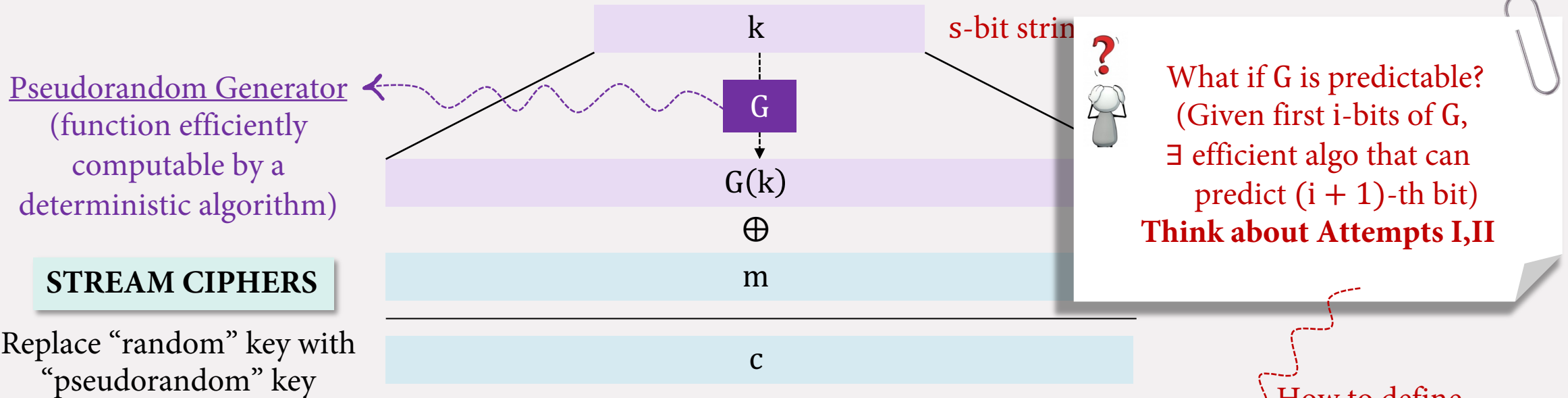
Need to define a different security notion!
We need some more basics –negligible functions, complexity theory



Perfect secure? **NO!**

How to overcome these limitations?

First limitation: Key as long as message! Let's try and fix using some process that expands the key..



NEXT
CLASS

Need to define a different security notion!
We need some more basics –negligible functions, complexity theory

DO NOT USE

Linear Congruential Generators

$$a, b \in \mathbb{Z}, p: \text{prime}$$

$$r[0] \equiv \text{seed}$$

$$\begin{array}{l} \rightarrow \\ r[i] \leftarrow a \cdot r[i-1] + b \bmod p \\ \text{output few bits of } r[i] \\ i++ \end{array}$$



HW: Read about **glibc random()**
Never use it!!
(Was used in Kerberos Version 4)

DO NOT USE RANDOM NUMBER GENERATORS LIKE THESE FOR CRYPTO!

How to overcome these limitations?

Relaxed

THREAT MODEL

How powerful is Eve (computationally)?

Bounded computation power
(Learn some complexity basics)

How can Eve attack the communication?

Other stronger attacks besides
ciphertext only attacks



Weaker
Eve

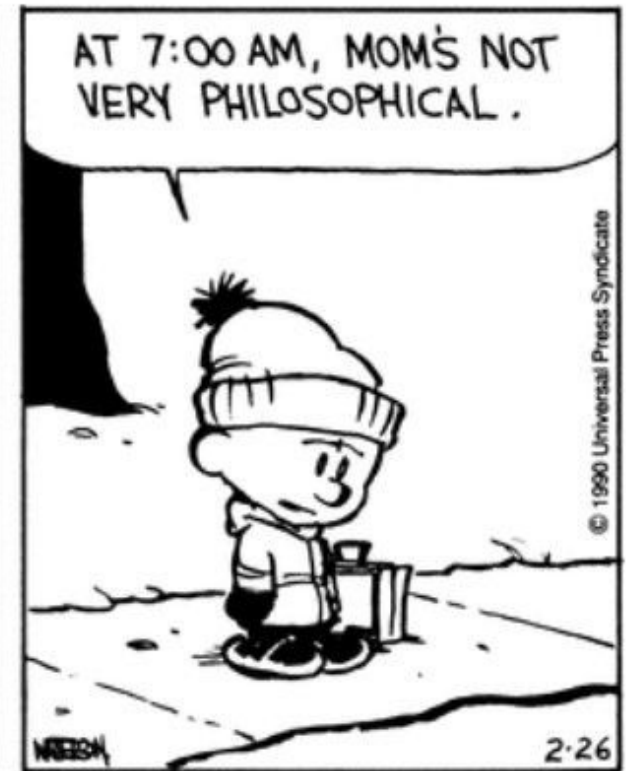
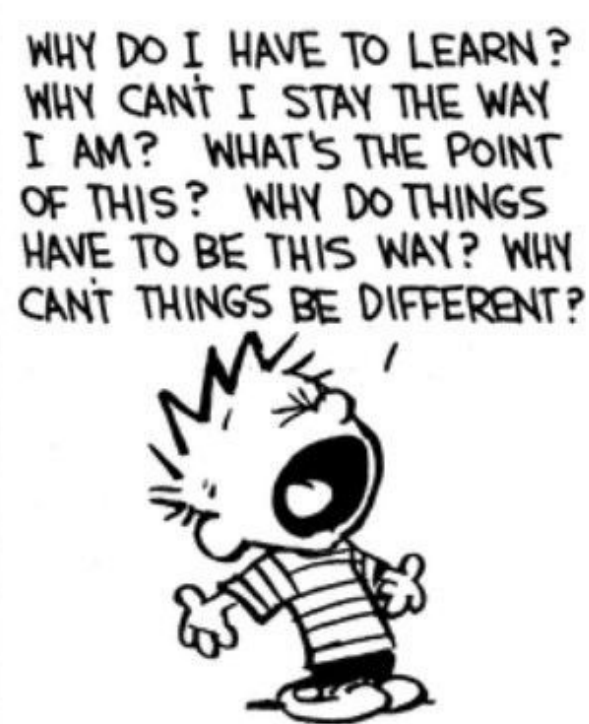
NEXT
CLASS

BREAK MODEL

Allow break with
some “small” probability

(What are negligible functions?)

Relaxed



THANK YOU